The Semyanistyi weighted-norm question in arXiv:1207.5180 was answered by arXiv:1210.5414

Literature-resolution packet for arXiv:1207.5180

13 August 2026

Status: literature already answered. Rubin's separate follow-up arXiv:1210.5414 proves the sharp weighted norm of the Semyanistyi fractional k-plane integral. Its formula has exactly the conjectured limit as $\alpha \to 0$. Banach-space adjoint duality gives the corresponding sharp result and norm limit for the dual operators.

1 The source question

Section 5, item 3 (printed page 14) of arXiv:1207.5180 considers

$$(R_k^\alpha f)(\tau) = \frac{1}{\gamma_{n-k}(\alpha)} \int_{\mathbb{R}^n} f(x)|x-\tau|^{\alpha+k-n}\,dx, \qquad \gamma_{n-k}(\alpha) = \frac{2^\alpha \pi^{(n-k)/2}\Gamma(\alpha/2)}{\Gamma((n-k-\alpha)/2)}.$$

It conjectures that the paper's sharp weighted-L^p method extends to these operators and their duals, and expects

$$\lim_{\alpha\to 0}\|R_k^\alpha\| = \|R_k\|$$

on the relevant weighted spaces. This is the complete mathematical content of item 3; items 1, 2, and 4 are different questions and are not classified by this packet.

2 The direct later theorem

Rubin's arXiv:1210.5414, submitted three months later, is titled *Weighted norm estimates for the Semyanistyi fractional integrals and Radon transforms*; it cites arXiv:1207.5180 explicitly. Its Theorem 3.2 (printed pages 8–9) states the sharp result for every $1 \le p \le \infty$ and $\alpha > 0$. With

$$\nu = \mu - \alpha - k/p', \qquad \alpha + k - n/p < \mu < n/p',$$

one has

$$\|R_k^\alpha f\|_{p,\nu}^{\sim} \le c_{\alpha,k}\|f\|_{p,\mu}, \qquad c_{\alpha,k} = \|R_k^\alpha\|,$$

where

$$c_{\alpha,k} = 2^{-\alpha}\pi^{k/2}\left(\frac{\sigma_{n-k-1}}{\sigma_{n-1}}\right)^{1/p} \times \frac{\Gamma((n/p'-\mu)/2)\Gamma((\mu+n/p-k-\alpha)/2)}{\Gamma((\mu+n/p)/2)\Gamma((n/p'-\mu+\alpha)/2)}.$$

Thus the later theorem supplies boundedness, the admissible power weights, and the exact operator norm, rather than only an estimate. Theorem 3.3 (printed page 12) records the zero-order theorem in the same notation:

$$\|R_k\| = c_k = \pi^{k/2} \left(\frac{\sigma_{n-k-1}}{\sigma_{n-1}} \right)^{1/p} \frac{\Gamma((\mu + n/p - k)/2)}{\Gamma((\mu + n/p)/2)}, \quad \nu = \mu - k/p'.$$

3 The asserted norm limit

Fix p and a weight exponent μ with

$$k - n/p < \mu < n/p'.$$

For all sufficiently small $\alpha > 0$, the fractional theorem applies with the compatible target weight $\nu_\alpha = \mu - \alpha - k/p'$. Continuity of $2^{-\alpha}$ and of Γ away from its poles gives

$$\begin{aligned} \lim_{\alpha \downarrow 0} c_{\alpha,k} &= \pi^{k/2} \left(\frac{\sigma_{n-k-1}}{\sigma_{n-1}} \right)^{1/p} \frac{\Gamma((n/p' - \mu)/2)\Gamma((\mu + n/p - k)/2)}{\Gamma((\mu + n/p)/2)\Gamma((n/p' - \mu)/2)} \\ &= c_k = \|R_k\|. \end{aligned}$$

This is precisely the expected limit, with the naturally varying target weight tending to $\mu - k/p'$. The upper restriction $\mu < n/p'$ is needed for the positive-order family; hence the limit is asserted on the common interior of the fractional weighted spaces, not at a parameter where R_k^α is unbounded.

4 Dual operators

The introduction of arXiv:1210.5414 explicitly says that the same approach applies to the dual Radon transforms and dual Semyanistyi integrals. More decisively, the dual conclusion follows immediately from its sharp primal theorem. Under the unweighted pairing of the two underlying measure spaces,

$$(R_k^\alpha)^* : L^q_\beta(V_{n,n-k} \times \mathbb{R}^{n-k}) \longrightarrow L^q_\delta(\mathbb{R}^n), \qquad \delta = \beta - \alpha - k/q,$$

is obtained by applying Theorem 3.2 with $p = q'$, $\mu = -\delta$, and $\nu = -\beta$. The admissible range becomes

$$\alpha - (n-k)/q < \beta < (n-k)/q',$$

and $\|(R_k^\alpha)^*\| = \|R_k^\alpha\|$ for the corresponding paired weights. Substitution into the displayed gamma formula gives the sharp dual constant. Letting $\alpha \downarrow 0$ cancels the same gamma pair and yields the sharp dual R_k^* norm in Theorem 1.1 of arXiv:1207.5180. Therefore the word “duals” in the source question creates no unresolved residue.

5 Conclusion and audit

The source’s item 3 and the later paper match in operator, weighted spaces, sharpness, and limiting statement. Because the supporting theorem was published as a separate paper by the same author and cites the source paper, this is a direct literature answer, not an independently inferred partial analogy.

Source audit. The official arXiv PDFs and source files of 1207.5180 and 1210.5414 were inspected. The source question is Section 5, item 3, printed page 14. The decisive later results are Theorems 3.2 and 3.3, printed pages 8–9 and 12. The target was absent from the run's cheap solution indexes before this packet.

Human review. Check the endpoint convention $\alpha \downarrow 0$ and the reparameterization in the one-line adjoint argument. The primal theorem identification and gamma-factor cancellation are literal.

References

[1] B. Rubin, *Weighted norm inequalities for k-plane transforms*, arXiv:1207.5180.

[2] B. Rubin, *Weighted norm estimates for the Semyanistyi fractional integrals and Radon transforms*, arXiv:1210.5414.